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Transcriptome Analysis of *Drosophila melanogaster* S2 Cells Under Amino-Acid Starvation

2.1. Introduction

Transcriptomics is the study of transcriptomes which represents the collection of RNA transcripts produced by a cell, tissue or an organism. It provides information about gene expression under a particular condition. Since gene expression changes under different conditions so transcriptome helps researchers understand how a gene responds to specific biological condition.

RNA seq is a widely used technique for transcriptomic analysis. It measures the expression of many genes at the same time. The sample genes under different conditions are compared to identify the change in gene expression. These genes are known as differentially expressed genes (DEGs).

During my internship I performed RNA seq analysis using Galaxy which provides a wide range of tools for the analysis of sequencing data. The dataset of raw sequencing reads of two replicates under the normal and starved condition was downloaded from NCBI. The main goal was to identify the differentially expressed genes. So, the data was first filtered to remove any duplication or contamination. The reads were then aligned with the reference genome followed by the quantification of reads. The genes that were differently expressed in both conditions were identified using DESeq2. The resulting genes were further analyzed using Gene Ontology Enrichment and KEGG enrichment to identify the pathways and the biological, cellular, and molecular functions that were enriched among the significant DEGs.

2.2. Objectives

- To analyze RNA-Seq data from *Drosophila melanogaster* under normal and amino acid starvation conditions.
- To check the quality of the raw sequencing data before further analysis.
- To identify genes that show significant differences in expression between the two conditions using DESeq2.

- To understand the biological functions of the differentially expressed genes using GO enrichment analysis.
- To identify the important pathways associated with the differentially expressed genes through KEGG enrichment analysis.
- To identify the interactions between the identified proteins and find important top 10 hub proteins using STRING and Cytoscape.

2.3. Materials and Methods

2.3.1. RNA-seq Dataset

Raw RNA-sequencing data for *Drosophila melanogaster* were obtained from the NCBI Sequence Read Archive (SRA) under study accession **SRP242243**. Four FASTQ datasets of two biological replicates under control and amino acid starved condition were selected.

Control: SRR10904051 and SRR10904052

Amino Acid Starvation: SRR10904053 and SRR10904054

The downloaded FASTQ files were used as the raw input for quality control, read trimming, and sequence alignment.

Table 1: Dataset Characteristics

Dataset Characteristic	Information
Organism	<i>Drosophila melanogaster</i>
Platform	Illumina
Instrument	NextSeq 500
Library layout	Single-end
Library selection	cDNA
Library source	Transcriptomic
SRA Study	SRP242243
Control samples	SRR10904051, SRR10904052
Amino Acid Starvation samples	SRR10904053, SRR10904054

2.3.2. Quality Control Using Falco

Quality control is the first step in any RNA-seq workflow, before performing alignment to ensure that only the high quality reads are used in analysis. Raw sequencing reads can contain low quality data and adapter contamination which is not good for the alignment accuracy and can lead to unreliable results in gene expression identification.

Falco was chosen to assess the quality of data over the more commonly used **FastQC** mainly for speed and efficiency. Falco performs same analysis as FASTQC but it is much faster than

FASTQC and uses less memory, which matters the most when working with multiple large files. The quality of the raw FASTQ reads was assessed using Falco across all four RNA-seq samples (SRR10904051, SRR10904052, SRR10904053, and SRR10904054) to check the quality of raw data.

2.3.3. Pre-processing of raw reads using Trimmomatic

Although the Falco quality control report had already shown good per-base quality and clean adapter content, trimmomatic was applied as a precautionary step so if there is any remaining low quality bases, they will be removed and to further improve the quality of reads for a better alignment. In this way, it improves the overall reliability and reproducibility of the data.

Trimmomatic is a widely used tool for trimming and filtering raw sequencing reads generated by Illumina platforms. It removes low-quality bases, adapter sequences (illuminaclip), and reads that fall below a defined length. In this analysis, Trimmomatic was used to filter the raw FASTQ files (SRR10904051, SRR10904052, SRR10904053, SRR10904054).

Table 2: Trimmomatic Parameters

Parameter	Value
Read type	Single-end
Adapter removal (ILLUMINACLIP)	Not performed
Trimming method	SLIDINGWINDOW
Window size	4 bases
Quality threshold	Q20
Quality encoding	Phred33

Alternative Tools and Justification for Using Trimmomatic

Other alternative tools for trimming include **fastp**, **Cutadapt**, and **Sickle**. fastp offers combined quality control and trimming. Cutadapt is mainly used for adapter specific trimming and has to combine with other tools for quality filtering, while Sickle performs only quality based trimming without having ability to remove adapters. Trimmomatic was chosen for this analysis because its parameters can be adjusted (e.g., sliding window size, quality threshold), which makes it a reliable and reproducible choice for this workflow.

2.3.4. Reference Genome and Read Alignment using HISAT2

Following the pre-processing, alignment was done using **HISAT2**. Alignment of reads to their corresponding genome is a crucial step because it determines the genomic location of reads. Alignment should be done accurately because poor alignment can lead to incorrect or incomplete mapping which produces biased or unreliable results.

For this purpose, HISAT2 was used to align the filtered reads to the reference genome. HISAT2 is a fast, memory-efficient, splice-aware alignment tool designed specifically for RNA-Seq data. Unlike standard aligners, it can correctly align reads that span exon–exon junctions (reads from spliced mRNA), making it suitable for mapping transcriptomic reads to a reference genome.

The reference genome of *Drosophila melanogaster* (**NCBI accession: GCF_000001215.4, RefSeq assembly**) was downloaded from NCBI and then used as an input for HISAT2 along with the trimmomatic result file. HISAT2 was then used to align the trimmed reads to the reference genome, in order to determine the genomic origin of each read.

Table 3: HISAT2 Parameters Used

Parameter	Value
Reference genome source	History (uploaded genome)
Library type	Single-end
Strand information	Unstranded
Input, alignment, scoring options	Default
Spliced alignment options	Default
Reporting options	Default

Alternative Tools

HISAT2 was preferred in this study for its balance of speed, low memory requirement, and high mapping accuracy for splice-aware alignment. Other commonly used alignment tools for RNA-seq data include:

- **STAR (Spliced Transcripts Alignment to a Reference)** with high accuracy and speed but it requires more memory than HISAT2.
- TopHat2 is also an alignment tool but is slow and less accurate than HISAT2.
- **Salmon** is also used for alignment which offers faster processing but does not generate BAM files.

2.3.5. Gene-Level Read Quantification

After aligning the reads to their genome positions, the next step was to count the number of reads corresponding to each gene. Quantification is a crucial step in RNA seq analysis as it forms the basis of differential expressed genes.

Gene quantification was performed using the featureCounts tool in galaxy. This tool is used to quantify gene expression by counting the number of sequencing reads that align with the annotated genomic features, such as genes or exons. It takes BAM file of aligned reads from HISAT2 along with a gene annotation file as an input and produces a count matrix representing

that how many reads were mapped to each gene in all the four samples. This count matrix is then used as an input file to determine the differentially expressed genes.

Table 4: *featureCounts* Parameters Used

Parameter	Value
Strand information	Unstranded
Annotation source	History (uploaded GFF)
Feature type	Exon
Gene identifier attribute	gene_id
Read type	Single-end
Minimum mapping quality	0
Multi-mapping reads	Counted (exon-exon junctions allowed)
Minimum overlap	1 base
Output format	Gene-ID + read-count (DESeq2/edgeR compatible)

Alternative Tools

featureCounts was preferred in this workflow for its better speed, simplicity. Other commonly used tools for gene-level quantification include

- **HTSeq-count** performs similar read counting but is slower than featureCounts on large datasets
- **RSEM**, which counts gene as well as isoform level expression but is more complex than featureCounts.
- **Salmon** is an alternative alignment tool as well as a gene quantification tool.

2.3.6. Differential Expression Analysis

After obtaining the gene count matrix, the next step was to identify the genes that showed a significant change in expression between the control and amino acid starvation conditions. This step helps to understand the biological impact of amino acid starvation.

DESeq2 within the galaxy was used for this purpose. DESeq2 is a widely used R/Bioconductor package for identifying differentially expressed genes from raw RNA seq count data. DESeq2 takes a gene level count matrix as input and performs steps such as normalization and dispersion estimation to determine which genes show significant changes in expression between experimental conditions. The gene count matrix obtained from featureCounts was used as input for DESeq2.

Raw gene counts were used as input values for DESeq2 instead of already normalized values such as **TPM/FPKM** because DESeq2 performs its own internal **normalization** as part of its statistical model, which requires raw count data to function correctly. Since different samples

can have different sequencing depths, DESeq2 performs normalization so that the genes can be compared fairly. It then performs the **dispersion** to measure the gene expression variation between biological replicates. Finally, genes that meet the selected **padj** (**<0.05**) and **log2FC** cutoffs are considered differentially expressed and are classified as either **upregulated** or **downregulated**.

Table 5: DESeq2 Parameters Used

Parameter	Value
Input data	Raw gene count matrix (from featureCounts)
Normalization method	Median-of-ratios (DESeq2 default)
Statistical test	Wald test
Comparison groups	Control vs. Amino Acid Starvation
Adjusted p-value (padj) cutoff	< 0.05
Log2 fold change cutoff	$ \log_2\text{FC} > 1$
Multiple testing correction	Benjamini-Hochberg (BH) method

Alternative Tools

DESeq2 was chosen for this study because it is reliable, widely trusted, and commonly used in RNA-seq research. Other tools commonly used for differential expression analysis include

- **edgeR** works in a similar way to DESeq2, but it calculates change in expression between samples slightly different than DESeq2.
- **limma-voom** was originally designed for microarray data but can also be used to work with RNA seq data.

2.3.7. Filtering of Differently Expressed Genes

After doing the expression analysis, the DESeq results were filtered to obtain the upregulated genes using the **Filter tool in Galaxy**. A condition of genes with $\text{padj} < 0.05$ and $\log_2\text{FoldChange} \geq 1$ was applied and a total of 560 upregulated genes were obtained and these genes were then used for further functional analysis.

2.3.8. Gene Ontology Enrichment Analysis

After identifying the upregulated genes from degs, GO enrichment analysis was performed to identify the biological functions associated with the upregulated genes. GO enrichment was performed using **GOEnrichment tool in Galaxy**. This tool works with the protein IDS of differentially expressed genes along with the **go-basic.obo** file and the protein annotation file of *Drosophila Melanogaster*.

The functional annotation was performed by using **eggNOG-mapper**. The protein FASTA sequence was downloaded from NCBI and then uploaded on eggNOG-mapper. The resulting annotation file from eggNOG-mapper contained the protein ids along with the GO term and KO (KEGG Orthology) identifiers. The annotation file was then filtered to remove the protein IDS that did not have the GO or KO annotations.

The go-basic.obo file was downloaded from the official website of Gene Ontology. After this the gene IDS of differentially expressed genes were converted into protein ids by using **R studio** and then all the three files (**go-basic.obo** as the database file, **annotation file** with protein ids and corresponding GO terms, and **protein IDS of upregulated genes** file as a study set file) were uploaded on GO enrichment tool.

The GO enrichment tool compares the study set file with the background gene set to identify the GO terms that were overrepresented among the degs.

Table 6: GOEnrichment Parameters Used

Parameter	Value
Multiple Test Correction	Benjamini-Hochberg
P-Value Cut-Off	0.01

The GOEnrichment terms were classified into three main categories:

- **Biological Process (BP)** describes the biological processes in which the DEGs were involved.
- **Molecular Function (MF)** describes the biological functions performed by the DEGs.
- **Cellular Component (CC)** describes the location of DEGs in cell.

2.3.9. KO (KEGG Orthology) Enrichment Analysis

The differentially expressed genes were further used for KEGG pathway enrichment. The KO identifiers from the annotated file for the upregulated genes were selected and processed in

The study set containing the protein IDS of 562 upregulated genes were loaded in R along with the background annotation file and then converted the protein IDS of study set into the KO ids.

KEGG Enrichment was done using clusterProfiler (R package) in R to check which biological pathways were overrepresented among the differentially expressed genes. Resulting p values were adjusted for multiple testing using Benjamini-Hochberg (BH) to remove the false positives and $pvalueCutoff = 0.05$ and $qvalueCutoff = 0.2$ were applied on the data to identify the significantly enriched pathways.

2.3.10. Protein–Protein Interaction (PPI) Network Analysis Using STRING

The differentially expressed genes were further analyzed to determine the protein-protein interactions among the proteins. For this purpose, the protein IDS of DEGs were uploaded on the official STRING website and the organism was selected as *Drosophila melanogaster*. An **interaction score of 0.700** was applied to obtain high-confidence interactions. STRING gives the known and predicted interactions among the proteins based on available evidence like experimental data.

2.3.11. Hub Gene Identification Using Cytoscape

The PPI network obtained from STRING was further analyzed using Cytoscape to identify the hub genes. The string network was imported in Cytoscape and cytohubba plugin was used to identify the central proteins that showed the highest number of interactions with other proteins. These genes were ranked according to their connectivity and the proteins with top interactions were selected as hub genes.

Hub genes are important because they have a large number of interactions with other proteins and these interactions show that they may have an important role in biological processes. However, the high connectivity with other proteins does not confirm that the protein is the most important, so the hub genes require further experimental validation.

2.4. Tools and Techniques Used

Table 7: Tools used in RNA seq analysis

Tool / Technique	Purpose
NCBI	Raw RNA-Seq sequencing data ko obtain karne ke liye use kiya gaya.
Galaxy	RNA-Seq data ki quality control, preprocessing, alignment, quantification aur downstream analysis ke liye use kiya gaya.
FastQC / Falco	Raw sequencing reads ki quality assess karne ke liye use kiya gaya.
Reference Genome	<i>Drosophila melanogaster</i> ke reads ko reference genome ke against align karne ke liye use kiya gaya.
Read Alignment	Sequencing reads ko reference genome ke corresponding regions ke saath match karne ke liye perform kiya gaya.
Read Quantification	Har gene ke corresponding reads ki number determine karne ke liye use kiya gaya.
DESeq2	Normal aur amino-acid starvation conditions ke darmiyan differentially expressed genes identify karne ke liye use kiya gaya.
Gene Ontology (GO) Enrichment	Differentially expressed genes ke biological processes, molecular functions aur cellular components ko understand karne ke liye use kiya gaya.
KEGG Enrichment Analysis	Differentially expressed genes se associated significantly enriched biological pathways identify karne ke liye use kiya gaya.

STRING	Protein–protein interaction network construct karne aur proteins ke functional associations study karne ke liye use kiya gaya.
Cytoscape	STRING se obtained interaction network ko visualize aur analyze karne ke liye use kiya gaya.
cytoHubba	Protein–protein interaction network mein important hub proteins identify karne ke liye use kiya gaya.

2.5. Results

2.5.1. Quality Control Results

The results obtained from Falco were examined to identify the quality issues in the raw sequencing data. The results are summarized in Table 2.

Table 8. Quality control results of raw FASTQ reads assessed using Falco.

Quality Metric	Result
Basic Statistics	Pass
Per Base Sequence Quality	Pass
Per Sequence Quality Scores	Pass
Per Base Sequence Content	Fail
Per Sequence GC Content	Warn
Per Base N Content	Pass
Sequence Length Distribution	Warn
Sequence Duplication Levels	Fail
Overrepresented Sequences	Warn
Adapter Content	Pass

The same Pass, Warn, and Fail results were observed across all four samples of data. Overall, raw data showed satisfactory per base and per sequence quality and there was no adapter contamination also. However fail and warn results were observed for other factors which were considered during the preprocessing step using trimmomatic.

Per base sequence quality Graph

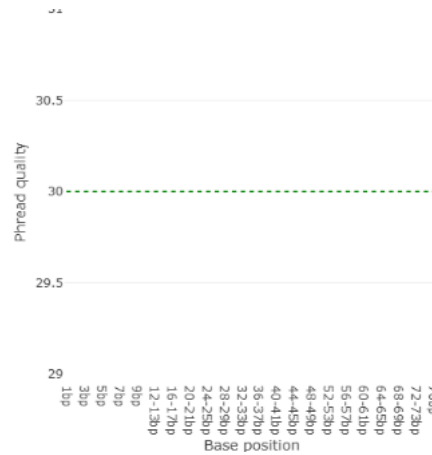


Figure 1 : Per-base sequence quality scores across read positions generated by Falco, showing consistently high Phred quality scores (Q30+) throughout the read length.

2.5.2. Read Trimming and Filtering Results

The raw reads were trimmed using trimmomatic and gives four trimmed files as a result.

Table 9: Trimmomatic Parameters

Parameter	Value
Read type	Single-end
Adapter removal (ILLUMINACLIP)	Not performed
Trimming method	SLIDINGWINDOW
Window size	4 bases
Quality threshold	Q20
Quality encoding	Phred33

```
@SRR10904051.1 1 length=76
ATTACACACATTGTCACTATTAC
+SRR10904051.1 1 length=76
????????????????????
@SRR10904051.2 2 length=76
CTTCTTGTCACCGCCGCTCGAAGT
+SRR10904051.2 2 length=76
????????????????????
@SRR10904051.3 3 length=74
ATACGAACGTCGAGCAACCGTGCCTGC
+SRR10904051.3 3 length=74
????????????????????
@SRR10904051.4 4 length=76
ATAAATAGTGCCAACTGCTACTTC
+SRR10904051.4 4 length=76
????????????????????
@SRR10904051.5 5 length=76
CAAACTGGCGTAAATAATCTAAAAATAA
+SRR10904051.5 5 length=76
????????????????????
```

Figure 2: Showing output of trimmomatic

2.5.3. Read Alignment Results

The trimmed reads from trimmomatic were aligned by using the HISAT2 tool which takes the trimmed files and the genome of organism as an input.

Genome assembly Release 6 plus ISO1 MT

reference

Download	datasets	API	FTP	Actions
NCBI RefSeq assembly	GCF_000001215.4			
Submitted GenBank assembly	GCA_000001215.4			
Taxon	Drosophila melanogaster (fruit fly)			
Synonym	dm6			
Assembly type	haploid			
Submitter	The FlyBase Consortium/Berkeley Drosophila Genome Project/Celera Genomics			
Date	Aug 1, 2014			

Figure 3: Reference Genome was downloaded from NCBI

Alignment Summary

Alignment summary of HISAT gives statistics for all four samples.

SRR10904051(control) <pre>HISAT2 summary stats: Total reads: 36981549 Aligned 0 time: 20110480 (54.38%) Aligned 1 time: 9809683 (26.53%) Aligned >1 times: 7061386 (19.09%) Overall alignment rate: 45.62%</pre>	SRR10904052 (control) <pre>HISAT2 summary stats: Total reads: 38276822 Aligned 0 time: 9573797 (25.01%) Aligned 1 time: 20295268 (53.02%) Aligned >1 times: 8407757 (21.97%) Overall alignment rate: 74.99%</pre>
SRR10904053 (amino acid starvation) <pre>HISAT2 summary stats: Total reads: 34092157 Aligned 0 time: 13105235 (38.44%) Aligned 1 time: 12550668 (36.81%) Aligned >1 times: 8436254 (24.75%) Overall alignment rate: 61.56%</pre>	SRR10904054 (amino acid starvation) <pre>HISAT2 summary stats: Total reads: 36107864 Aligned 0 time: 10585054 (29.32%) Aligned 1 time: 17207127 (47.65%) Aligned >1 times: 8315683 (23.03%) Overall alignment rate: 70.68%</pre>

Figure 4: Alignment summary statistics from HISAT2

2.5.4. Gene-Level Quantification Results

featureCounts was run separately on each of the four aligned BAM files, producing an individual count file for each sample. The results of 10 genes were then summarized in a table to show the difference in the read counts among the same genes in all four samples. Due to which some of the genes were highly expressed

while other genes were less or not expressed. (Table6).

59: featureCounts on dataset 43 and 55 control: Counts

ok

Preview Visualize Details Edit

Geneid	HISAT2 on dataset 25 and 54: aligned reads (BAM)
Dmel_CR40469	6954
Dmel_CG17636	0
Dmel_CG40494	19
Dmel_CR43552	0
Dmel_CG17131	40
Dmel_CR45473	59
Dmel_CG17707	7
Dmel_CR43863	0
Dmel_CG3038	104
Dmel_CG2995	611
Dmel_CG42376	100

Figure 5: Output data of SRR10904051(control)

Table 10: Combined featureCounts Output (Sample Genes, All 4 Samples)

Gene ID	SRR10904051(control)	SRR10904052 (control)	SRR10904053 (amino acid starvation)	SRR10904054 (amino acid starvation)
Dmel_CR40469	6954	39350	7666	14994
Dmel_CG17636	0	0	0	0
Dmel_CG40494	19	36	20	24
Dmel_CR43552	0	0	0	0
Dmel_CG17131	40	83	29	40
Dmel_CR45473	59	213	35	68
Dmel_CG17707	7	47	16	13
Dmel_CR43863	0	3	0	2
Dmel_CG3038	104	196	65	68
Dmel_CG2995	611	1111	537	671

Table 11: featureCounts Summary Statistics

Status	SRR10904051(control)	SRR10904052 (control)	SRR10904053 (amino acid starvation)	SRR10904054 (amino acid starvation)
Assigned	8,689,945	18,276,354	11,065,292	15,382,924
Unassigned_Unmapped	20,110,480	9,573,797	13,105,235	10,585,054
Unassigned_MultiMapping	32,111,776	36,755,023	38,783,757	36,595,362
Unassigned_NoFeatures	642,652	1,156,596	856,457	1,149,212
Unassigned_Ambiguity	477,086	862,318	628,919	674,991

Unassigned_Read_Type / Singleton / MappingQuality / Chimera / FragmentLength / Duplicate / Secondary / NonSplit / Overlapping Length	0 (all)	0 (all)	0 (all)	0 (all)
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2.5.5. Differential Expression Analysis Results

The files from featureCounts were uploaded on DESeq2 and it produces different graphs as a result.

Principal Component Analysis (PCA)

PCA was used to see the overall variation among the samples and to check whether the samples from the same condition showed similar expression patterns. The PCA plot showed the distribution of the Control and Amino Acid Starvation samples based on their overall gene expression. Samples belonging to the same condition were expected to cluster together, while separation between the two groups indicated differences in their overall expression profiles.



Figure 6. PCA plot showing clear separation between Control and Amino Acid Starvation groups along PC1.

MA-Plot

The MA-plot (Figure 9) shows the relationship between mean expression and log fold change for each gene, with statistically significant genes highlighted in blue. A large number of genes were found to be both upregulated and downregulated under starvation conditions, indicating a transcriptional response to amino acid starvation.

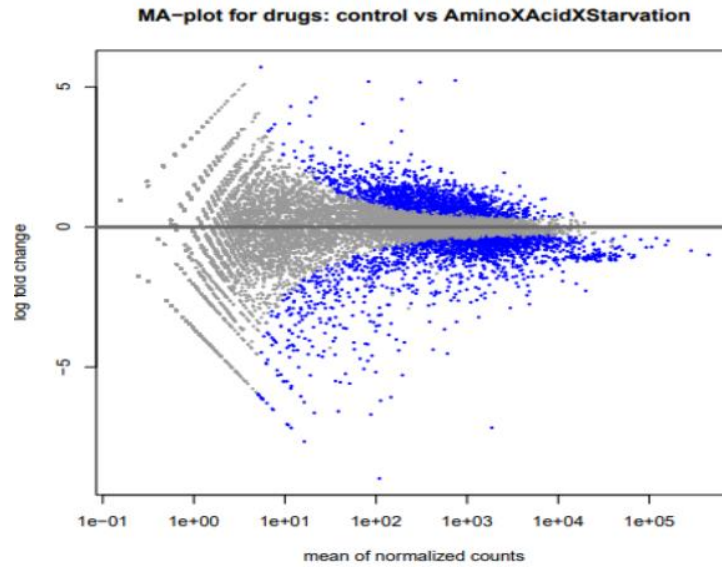


Figure 7. MA-plot showing log fold change versus mean expression for Control vs Amino Acid Starvation, with significant genes ($p_{adj} < 0.05$) shown in blue.

2.5.6. Gene Ontology Enrichment

GO enrichment was performed on upregulated genes to identify the biological, molecular and cellular processes in which the upregulated genes were involved.

Biological Process

The enriched biological processes were mainly related to mitochondrial functions, including mitochondrion organization, sperm mitochondrion organization, and mitochondrial translation. Other enriched processes were related to cell division and meiosis, such as cell division, organelle fission, meiotic cell cycle checkpoint signaling, and fusion of sperm to egg plasma membrane.

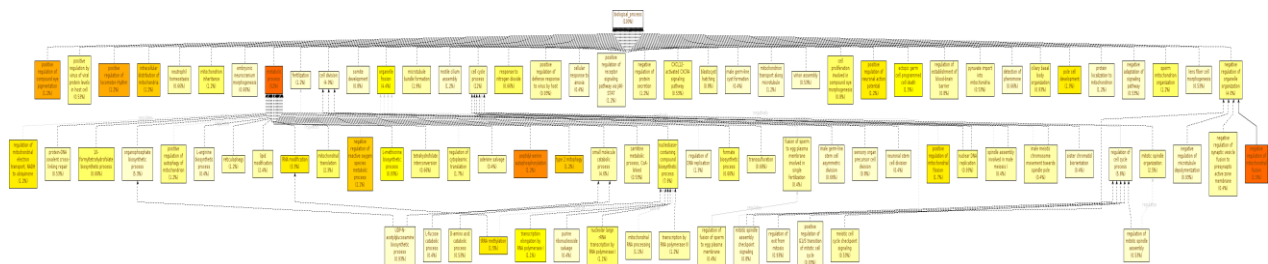
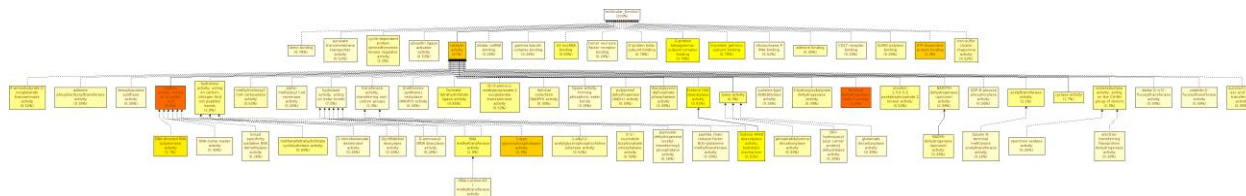


Figure 8. GO enrichment analysis of upregulated genes showing enriched terms in biological process.

The enriched molecular functions were mainly related to catalytic activities. These included catalytic activity acting on a nucleic acid, DNA-directed RNA polymerase activity, RNA methyltransferase activity, and histone H4K/H4K8 deacetylase activity.



Cellular Component

The GO analysis showed that many of the upregulated genes were associated with mitochondria. The main enriched terms included mitochondrion (15%), mitochondrial membrane (8.5%), mitochondrial matrix (5.8%), and mitochondrial outer membrane (3.8%). Other enriched cellular components included the germline-specific structure Nebenkern (1.5%) and the DNA-directed RNA polymerase complex (2.0%).

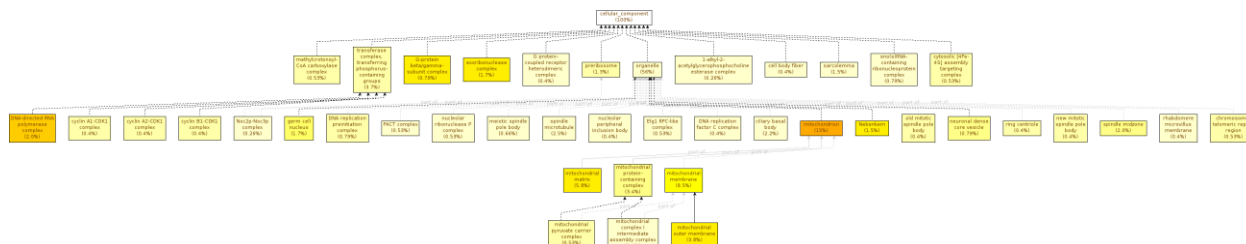


Figure 10. GO enrichment analysis of upregulated genes showing enriched terms in cellular component categories.

2.5.7. KEGG Pathway Enrichment Results

KEGG Enrichment was done using R programming to identify the biological pathways that were overrepresented within the upregulated genes.

Significantly Enriched Pathways

KEGG pathway enrichment analysis identified **nine significantly enriched pathways** among the upregulated genes (Figure X). The Motor proteins pathway showed the highest enrichment, with approximately 25 genes and a p.adjust value of around 0.01. DNA replication and RNA polymerase were also enriched, with around 10–11 genes involved in each pathway. The Cell

cycle pathway had 16 genes. Other significantly enriched pathways included Oocyte meiosis, progesterone-mediated oocyte maturation, p53 signaling pathway, cytosolic DNA-sensing pathway and beta-Alanine metabolism, with around 6–11 genes involved in these pathways.

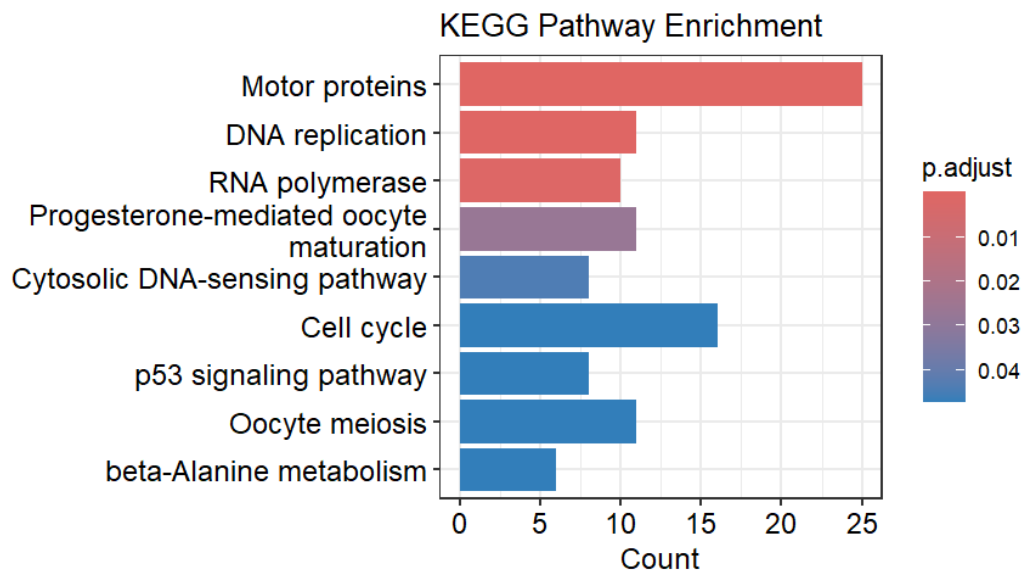


Figure 11. KEGG pathway enrichment analysis of upregulated genes showing the significantly enriched pathways and the number of genes associated with each pathway.

2.5.8. Protein–Protein Interaction (PPI) Network Analysis

The STRING analysis generated a PPI network containing **464 proteins and 1832 interactions**. A minimum interaction score of **0.700** was used to retain high confidence interactions. The interaction scores ranged from **0.700 to 0.999**, with an average score of **0.8603**, indicating that the network mainly consisted of high-confidence protein interactions.

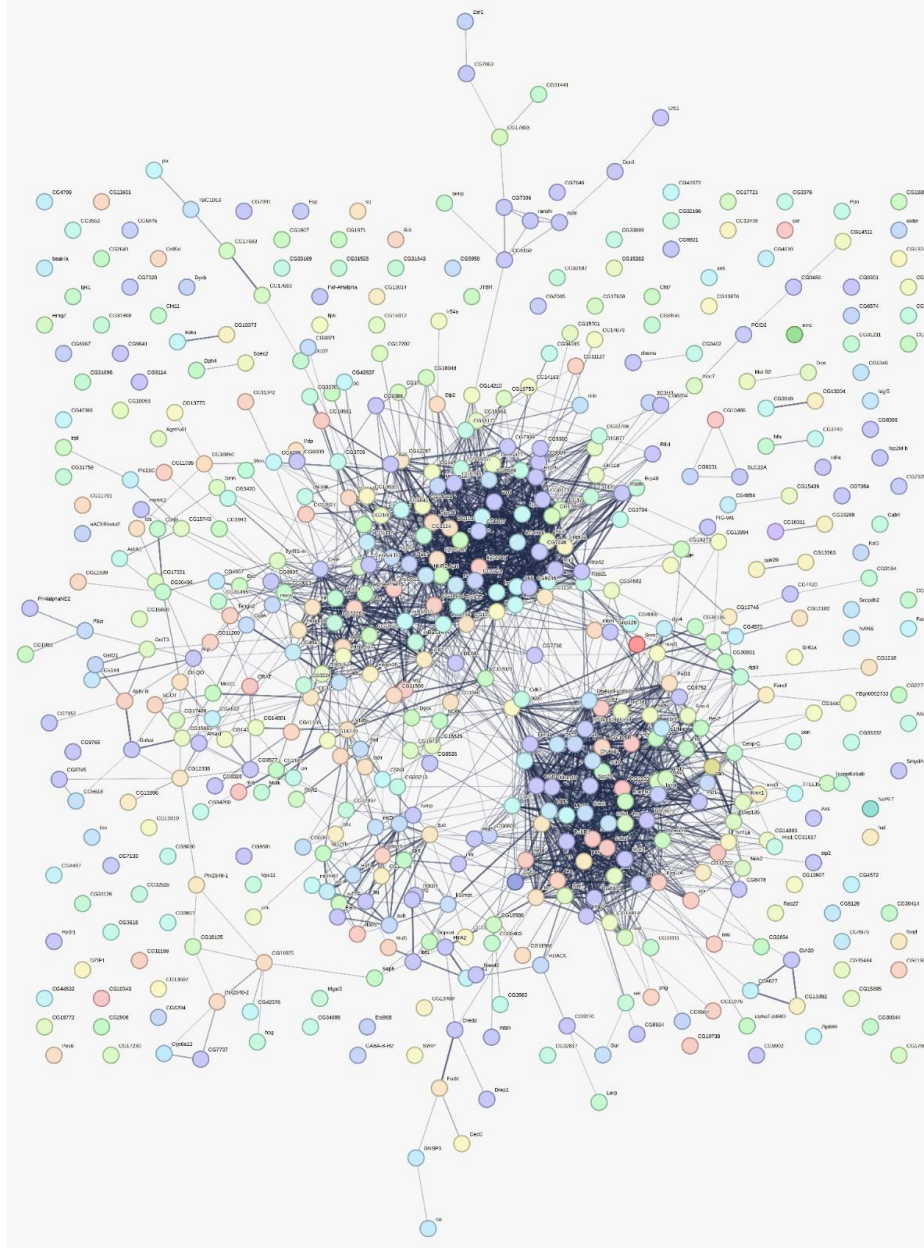


Figure 12. Protein–protein interaction network of the upregulated proteins generated using STRING with a minimum interaction score of 0.700.

2.5.9. Hub Gene Identification

The STRING interaction network was further analyzed in Cytoscape using the **CytoHubba** plugin to identify the most highly connected proteins. The **Maximum Clique Centrality (MCC)** method was used to rank the proteins according to their network connectivity. The top 10 proteins were selected as hub genes. **CG11837** ranked first with an MCC score of 1.01×10^7 . Other top hubgenes are shown in the table below:

Table 11: top 10 hubgenes

Rank	Hub gene	MCC score
1	CG11837	10,120,486
2	CG7246	9,825,272
3	wcd	9,753,254
4	CG1789	9,035,314
5	CG6937	6,406,504
6	Sas10	6,307,944
7	NHP2	6,302,701
8	CG9246	6,056,887
9	l(2)05287	5,733,216
10	CG4806	5,249,736

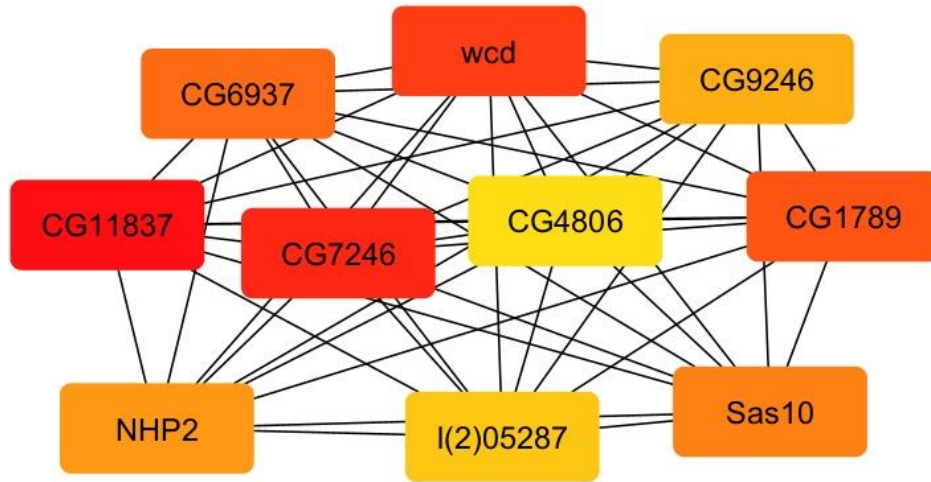


Figure 13. Top 10 hub proteins identified from the STRING PPI network using the MCC method in CytoHubba.

2.6. Discussion

Differential Expression Analysis

The differential expression analysis showed that amino acid starvation caused changes in gene expression compared with the control condition. A number of genes showed increased expression under the starvation condition, indicating that the cells responded to the lack of amino acids. These changes suggest that the cells may activate different mechanisms to adjust to the change in nutrient availability so that they can survive. The upregulated genes were therefore

selected for further analysis to understand the biological processes and pathways involved in this response.

Gene Ontology Enrichment Analysis

The GO enrichment results showed that many of the upregulated genes were associated with mitochondrial components and functions. Several enriched terms were related to the mitochondrion, mitochondrial membrane, mitochondrial matrix, and mitochondrial organization. This may indicate that amino acid starvation affects mitochondrial activity and related cellular functions. Since mitochondria are involved in energy production and metabolism, changes in mitochondrial-related genes may help cells adjust to limited nutrient availability. Other enriched terms related to cell division and meiotic processes also suggest that starvation may influence growth of and reproductive functions.

KEGG Pathway Enrichment Analysis

The KEGG analysis showed enrichment of pathways involved in different cellular and metabolic activities. Pathways such as DNA replication, RNA polymerase, cell cycle, p53 signaling, and cytosolic DNA sensing were among the enriched pathways. The enrichment of DNA replication and cell cycle pathways may indicate changes in cellular growth and division under amino acid starvation. Similarly, the involvement of the p53 signaling pathway may be related to the cellular response to stress. The enrichment of beta Alanine metabolism also suggests that amino acid starvation may affect metabolic processes as the cells respond to changes in nutrient availability. Overall, these findings indicate that the response to amino acid starvation involves several cellular pathways rather than only one specific process.

Protein–Protein Interaction Network Analysis

The STRING analysis showed that many of the proteins associated with the upregulated genes were connected with each other. The network contained 464 proteins and 1,832 interactions at a high-confidence interaction score of 0.700. This indicates that the proteins may work together in different biological processes affected by amino acid starvation. Looking at the interactions between proteins provides additional information because changes in one gene may be related to other proteins involved in the same cellular functions. The PPI network therefore helped to identify groups of proteins that may be involved in the overall response.

Hub Gene Analysis

The hub gene analysis identified highly connected proteins within the PPI network using the MCC method in CytoHubba. The top-ranked proteins included CG11837, CG7246, wcd, CG1789, CG6937, Sas10, NHP2, CG9246, l(2)05287, and CG4806. These proteins showed high connectivity within the network, suggesting that they may have important roles in the

interactions among the affected proteins. The hub genes can therefore be considered potential key proteins for further investigation. However, their high connectivity does not by itself prove that they are responsible for the starvation response, so further experimental studies would be needed to confirm their specific roles.

Overall Interpretation

Overall, the results show that amino acid starvation affects different cellular activities in *Drosophila melanogaster*. The GO analysis mainly showed changes related to mitochondrial functions, while the KEGG analysis highlighted pathways involved in metabolism, DNA replication, transcription, cell cycle, and stress response. The STRING analysis also showed that many of these proteins interact with each other, and the hub gene analysis helped identify some highly connected proteins. Overall, these results suggest that the cells respond to amino acid starvation by changing several metabolic and cellular processes at the same time.

2.7. Conclusion

In this study, RNA-Seq data was analyzed to see how amino acid starvation affects gene expression in *Drosophila melanogaster*. Differential expression analysis was used to find the genes that changed under starvation conditions. The upregulated genes were then analyzed using GO and KEGG enrichment analysis to understand their functions and pathways. The analysis showed that these genes were involved in different processes such as metabolism, mitochondrial functions, DNA replication, and cell cycle. STRING and Cytoscape were also used to study protein interactions and find important hub proteins. Overall, the results showed that amino acid starvation affects several cellular and metabolic processes in *Drosophila melanogaster*.

2.8. Limitations

One limitation of this study was that the analysis was based on available RNA-seq data and computational methods. The study included only a limited number of biological replicates (two), which may affect the reliability of some statistical results. The GO, KEGG, and PPI analyses were also based on available annotations and predicted or known interactions. Therefore, the identified pathways and hub genes provide useful information but would need further experimental validation to confirm their biological roles.

